Advanced Computer-Aided VLSI System Design
Homework 4 Report

Due Tuesday, May 26, 13:59

Student ID: B11901063

Student Name: 李佑倫

APR Results

1. Fill in the blanks below.

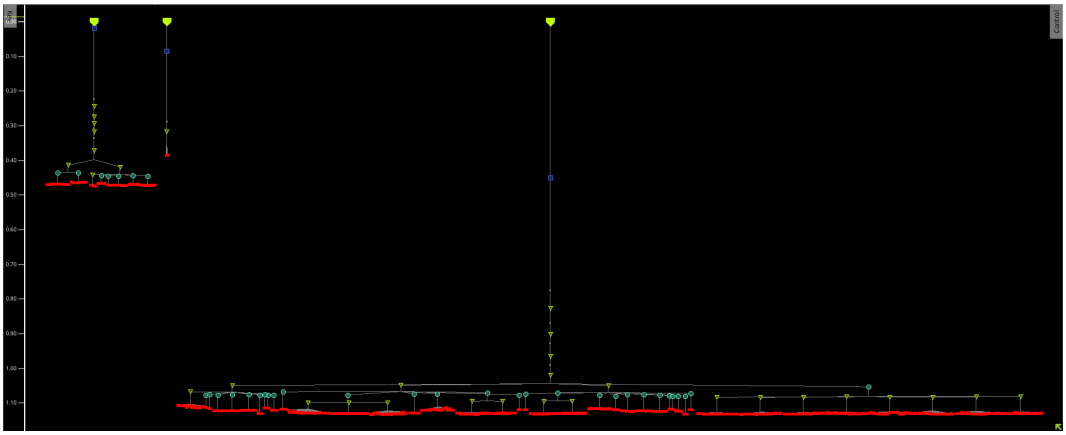
Design Stage	Description	Value
P&R	Number of DRC violations (ex: 0) (Verify -> Verify Geometry…)	0
	Number of LVS violations (ex: 0) (Verify -> Verify Connectivity…)	0
	Die Area (um^2)	2545090.57
	Core Area (um^2)	1475743.90
What type of APR design are you working on? Chip-level design: Netlist with I/O pads Block design: Netlist without I/O pads		Chip-level design

Questions and Discussion

1. Attach the snapshot of scan chain status (command: report_scan_chain) (5%).

```
*** Scan Trace Summary (runtime: cpu: 0:00:00.0 , real: 0:00:00.0):
Successfully traced 3 scan chains (total 1966 scan bits).
Start applying DEF ordered sections ...
Successfully applied all DEF ordered sections.
*** Scan Sanity Check Summary:
*** 3 scan chains passed sanity check.
save out the reporting file "scanChain.rpt"
```

2. Attach the snapshot of CCOpt Clock Tree Debugger result (5%).



3. Attach the snapshot of DRC and LVS checking after routing. (5%)

```
innovus 33> verify_drc
#-check_ndr_spacing auto                # enums={true false auto}, default=auto,  user setting
#-check_same_via_cell true              # bool, default=false, user setting
#-report top.drc.rpt                    # string, default="", user setting
 *** Starting Verify DRC (MEM: 18489.6) ***

  VERIFY DRC ..... Starting Verification
  VERIFY DRC ..... Initializing
  VERIFY DRC ..... Deleting Existing Violations
  VERIFY DRC ..... Creating Sub-Areas
  VERIFY DRC ..... Using new threading
 VERIFY DRC ..... Sub-Area: {-3.696 -60.276 24.576 -32.004} 3 of 3249  Thread : 2
 VERIFY DRC ..... Sub-Area: {222.480 -60.276 250.752 -32.004} 11 of 3249  Thread : 5
 VERIFY DRC ..... Sub-Area: {-60.240 -60.276 -31.968 -32.004} 1 of 3249  Thread : 7
 VERIFY DRC ..... Sub-Area: {279.024 -60.276 307.296 -32.004} 13 of 3249  Thread : 2
 VERIFY DRC ..... Sub-Area: {392.112 -60.276 420.384 -32.004} 17 of 3249  Thread : 5
 VERIFY DRC ..... Sub-Area: {-31.968 -60.276 -3.696 -32.004} 2 of 3249  Thread : 7
 VERIFY DRC ..... Sub-Area: {335.568 -60.276 363.840 -32.004} 15 of 3249  Thread : 3
 VERIFY DRC ..... Sub-Area: {52.848 -60.276 81.120 -32.004} 5 of 3249  Thread : 4
 VERIFY DRC ..... Sub-Area: {250.752 -60.276 279.024 -32.004} 12 of 3249  Thread : 2
 VERIFY DRC ..... Sub-Area: {24.576 -60.276 52.848 -32.004} 4 of 3249  Thread : 4
```

```
 VERIFY DRC ..... Sub-Area: {1240.272 1438.140 1268.544 1466.412} 3068 of 3249  Thread : 2
 VERIFY DRC ..... Sub-Area: {1296.816 1381.596 1325.088 1409.868} 2956 of 3249  Thread : 1
 VERIFY DRC ..... Sub-Area: {1268.544 1409.868 1296.816 1438.140} 3012 of 3249  Thread : 1
 VERIFY DRC ..... Sub-Area: {1381.632 1381.596 1409.904 1409.868} 2959 of 3249  Thread : 6
 VERIFY DRC ..... Sub-Area: {1325.088 1409.868 1353.360 1438.140} 3014 of 3249  Thread : 2
 VERIFY DRC ..... Sub-Area: {81.120 1381.596 109.392 1409.868} 2913 of 3249  Thread : 0
 VERIFY DRC ..... Sub-Area: {1296.816 1409.868 1325.088 1438.140} 3013 of 3249  Thread : 1
 VERIFY DRC ..... Sub-Area: {1409.904 1381.596 1438.176 1409.868} 2960 of 3249  Thread : 6
 VERIFY DRC ..... Sub-Area: {1409.904 1409.868 1438.176 1438.140} 3017 of 3249  Thread : 6
 VERIFY DRC ..... Sub-Area: {81.120 1409.868 109.392 1438.140} 2970 of 3249  Thread : 0
 VERIFY DRC ..... Sub-Area: {1353.360 1381.596 1381.632 1409.868} 2958 of 3249  Thread : 2
 VERIFY DRC ..... Sub-Area: {1353.360 1409.868 1381.632 1438.140} 3015 of 3249  Thread : 2
 VERIFY DRC ..... Thread : 0 finished.
 VERIFY DRC ..... Thread : 1 finished.
 VERIFY DRC ..... Thread : 6 finished.
 VERIFY DRC ..... Sub-Area: {1381.632 1409.868 1409.904 1438.140} 3016 of 3249  Thread : 2
 VERIFY DRC ..... Thread : 2 finished.

  Verification Complete : 0 Viols.

 *** End Verify DRC (CPU: 0:08:50  ELAPSED TIME: 69.00  MEM: 0.0M) ***
```

```
******** Start: VERIFY CONNECTIVITY ********
Start Time: Sun May 17 16:09:28 2026

Design Name: top
Database Units: 2000
Design Boundary: (-60.2400, -60.2760) (1535.1000, 1535.0520)
Error Limit = 1000; Warning Limit = 50
Check all nets
Use 8 pthreads
*** 16:09:28 *** Building data for Net VDD
*** 16:09:28 *** Building data for Net VSS
*** 16:09:28 *** Building data for Net VSS
*** 16:09:28 *** Building data for Net VDD
```

```
*** 16:09:29 *** Building data for Net VDD
*** 16:09:29 *** Building data for Net VSS
*** 16:09:29 *** Building data for Net VDD
*** 16:09:29 *** Building data for Net VSS
*** 16:09:29 *** Building data for Net VDD
*** 16:09:29 *** Building data for Net VSS
*** 16:09:29 *** Building data for Net VDD
*** 16:09:29 *** Building data for Net VSS
*** 16:09:29 *** Building data for Net VDD
*** 16:09:29 *** Building data for Net VSS
*** 16:09:29 *** Building data for Net VDD
*** 16:09:30 *** Building data for Net VSS
*** 16:09:30 *** Building data for Net VDD

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Sun May 17 16:09:44 2026
Time Elapsed: 0:00:16.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:32.1  MEM: -1716.000M)
```

4. Attach the snapshot of the timing report for **setup time and hold time** with no timing violation (post-route). (5%)

```
------------------------------------------------------------------
         timeDesign Summary
------------------------------------------------------------------

Setup views included:
 AV_func_0p72v_m40c AV_func_0p72v_125c

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg |reg2cgate| in2reg  | reg2out | in2out  | default |
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.549  |  1.730  |  1.760  |  1.944  |  0.549  |   N/A   |  0.000  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |    0    |   N/A   |    0    |
|         All Paths:|  1944   |  1859   |   35    |   21    |   49    |   N/A   |    0    |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |             Real              |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+
```

```
------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

Hold views included:
 AV_func_0p72v_m40c AV_func_0p72v_125c AV_func_0p88v_m40c
 AV_func_0p88v_125c AV_scan_0p88v_m40c AV_scan_0p88v_125c

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg |reg2cgate| in2reg  | reg2out | in2out  | default |
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.005  |  0.005  |  0.006  |  0.008  |  0.755  |   N/A   |  0.000  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |    0    |   N/A   |    0    |
|          All Paths:|  3909   |  3823   |   35    |   22    |   49    |   N/A   |    0    |
+--------------------+---------+---------+---------+---------+---------+---------+---------+
```

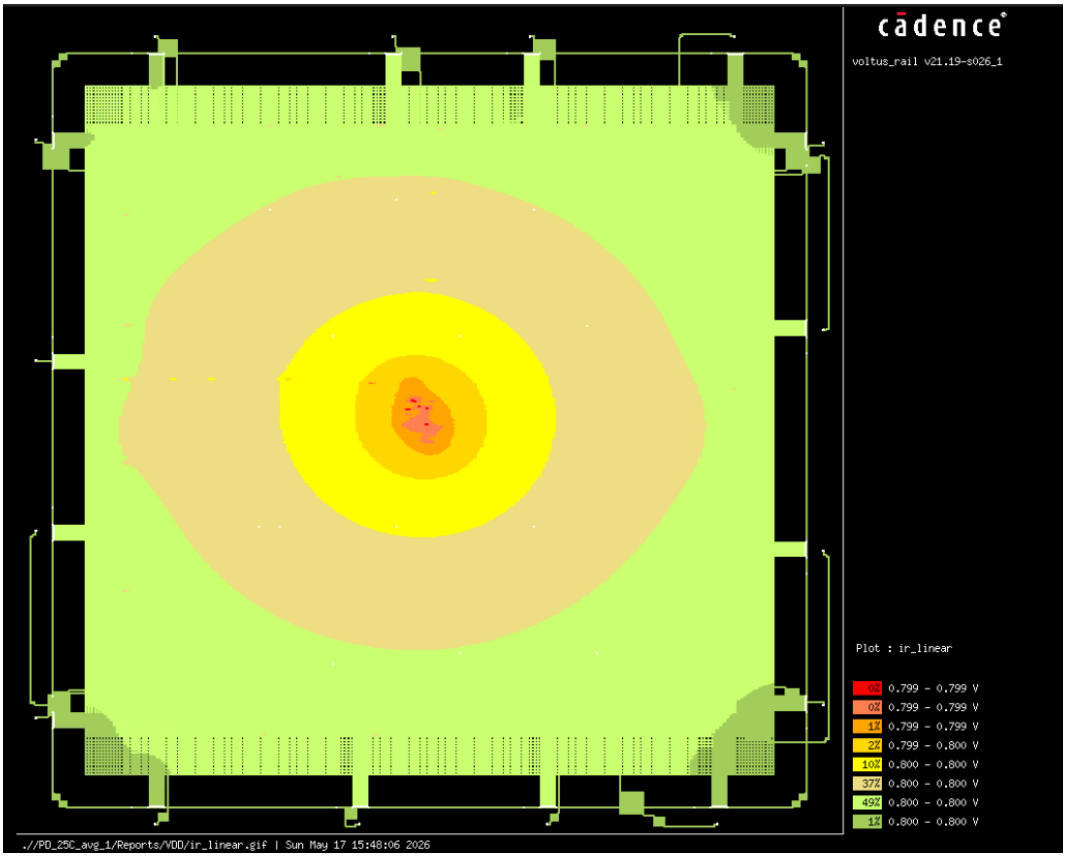
5. Show the critical path after post-route optimization. What is the path type? (5%)
   Through command：**`report_timing -max_path 1`**
   (The slack of the critical path should match the smallest slack in the timing report)

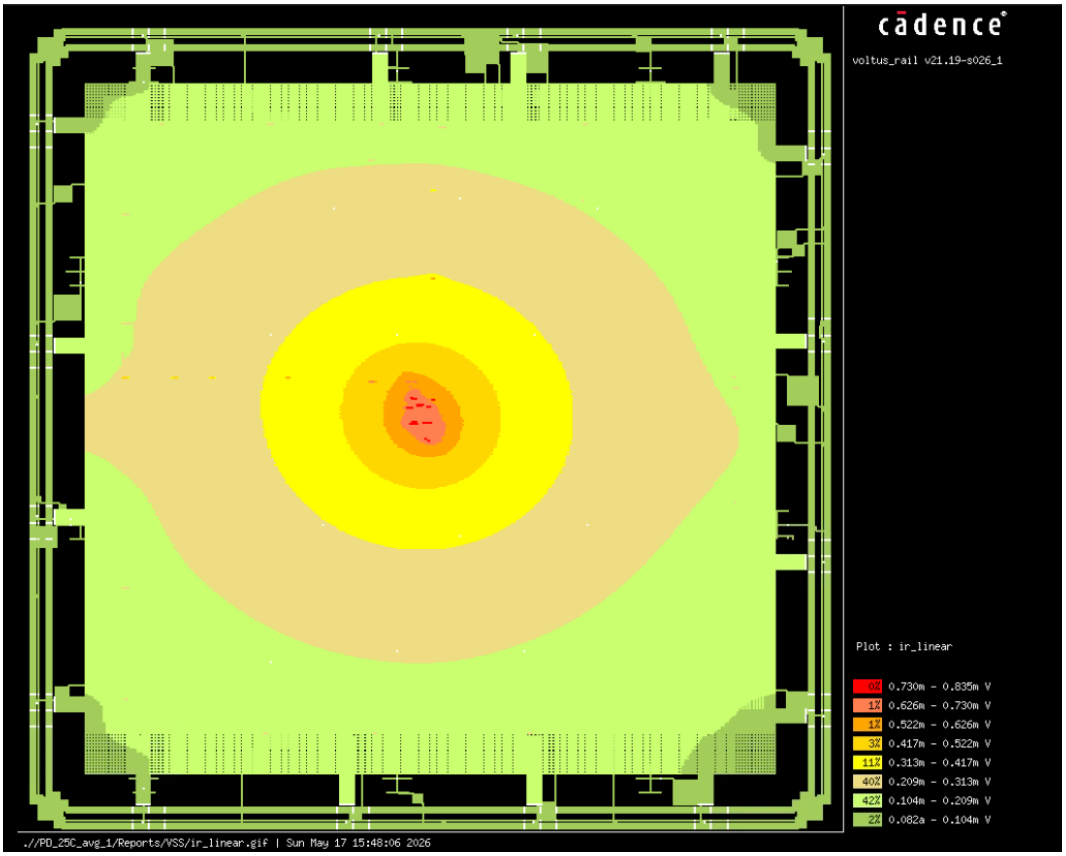
```
Path 1: MET Late External Delay Assertion
Endpoint:   ARVALID                                (^) checked with
leading edge of 'clk_ram'
Beginpoint: read_ctrl0_AXI_controller0_r_state_reg_1_/Q (v) triggered by
leading edge of 'clk_ram'
Path Groups: {reg2out}
Analysis View: AV_func_0p72v_125c
Other End Arrival Time          0.000
- External Delay                0.000
+ Phase Shift                   2.500
+ CPPR Adjustment               0.000
- Uncertainty                   0.100
= Required Time                 2.400
- Arrival Time                  1.851
= Slack Time                    0.549
     Clock Rise Edge                     0.000
     + Clock Network Latency (Prop)      0.574
     = Beginpoint Arrival Time           0.574
     +-----------------------------------------------------------------------------------------------------------+
     |                  Instance                  |     Arc      |       Cell        | Delay | Arrival | Required |
     |                                            |              |                   |       |  Time   |   Time   |
     |--------------------------------------------+--------------+-------------------+-------+---------+----------|
     | read_ctrl0_AXI_controller0_r_state_reg_1_  | CP ^         |                   |       |   0.574 |    1.123 |
     | read_ctrl0_AXI_controller0_r_state_reg_1_  | CP ^ -> Q v  | SDFQD2BWP20P90    | 0.084 |   0.658 |    1.206 |
     | FE_PHC2451_read_ctrl0_AXI_controller0_r_state_1 | I v -> Z v   | DEL050D1BWP20P90  | 0.090 |   0.748 |    1.296 |
     | FE_PHC1197_read_ctrl0_AXI_controller0_r_state_1 | I v -> Z v   | DEL025D1BWP20P90  | 0.051 |   0.798 |    1.347 |
     | U5280                                      | B1 v -> ZN ^ | INR2D1BWP16P90LVT | 0.109 |   0.907 |    1.456 |
     | FE_OFC107_n705                             | I ^ -> ZN v  | INVD1BWP16P90LVT  | 0.135 |   1.042 |    1.591 |
     | FE_OFC108_n705                             | I v -> ZN ^  | INVD1BWP16P90LVT  | 0.141 |   1.183 |    1.732 |
     | o_pad_ARVALID                              | I ^ -> PAD ^ | PDCDG_H           | 0.668 |   1.851 |    2.400 |
     |                                            | ARVALID ^    |                   | 0.000 |   1.851 |    2.400 |
     +-----------------------------------------------------------------------------------------------------------+
```

6. Attach the snapshot of GDS stream out messages. (5%)

```
Scanning GDS file /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/GDS/N16ADFP_StdCell.gds to register cell name ......
Scanning GDS file /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdio/N16ADFP_StdIO/GDS/N16ADFP_StdIO.gds to register cell name ......
Scanning GDS file /share1/tech/ADFP/Executable_Package/Collaterals/IP/bondpad/N16ADFP_BondPad/GDS/N16ADFP_BondPad.gds to register cell name ......
Scanning GDS file /share1/tech/ADFP/Lab_adfp/library/memory/rf_2p_hse.gds to register cell name ......
Merging GDS file /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/GDS/N16ADFP_StdCell.gds ......
        ****** Merge file: /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/GDS/N16ADFP_StdCell.gds has version number: 600.
        ****** Merge file: /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/GDS/N16ADFP_StdCell.gds has units: 1000 per micron.
        ****** unit scaling factor = 1 ******
Merging GDS file /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdio/N16ADFP_StdIO/GDS/N16ADFP_StdIO.gds ......
        ****** Merge file: /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdio/N16ADFP_StdIO/GDS/N16ADFP_StdIO.gds has version number: 5.
        ****** Merge file: /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdio/N16ADFP_StdIO/GDS/N16ADFP_StdIO.gds has units: 1000 per micron.
        ****** unit scaling factor = 1 ******
Merging GDS file /share1/tech/ADFP/Executable_Package/Collaterals/IP/bondpad/N16ADFP_BondPad/GDS/N16ADFP_BondPad.gds ......
        ****** Merge file: /share1/tech/ADFP/Executable_Package/Collaterals/IP/bondpad/N16ADFP_BondPad/GDS/N16ADFP_BondPad.gds has version number: 5.
        ****** Merge file: /share1/tech/ADFP/Executable_Package/Collaterals/IP/bondpad/N16ADFP_BondPad/GDS/N16ADFP_BondPad.gds has units: 1000 per micron.
        ****** unit scaling factor = 1 ******
Merging GDS file /share1/tech/ADFP/Lab_adfp/library/memory/rf_2p_hse.gds ......
        ****** Merge file: /share1/tech/ADFP/Lab_adfp/library/memory/rf_2p_hse.gds has version number: 5.
        ****** Merge file: /share1/tech/ADFP/Lab_adfp/library/memory/rf_2p_hse.gds has units: 1000 per micron.
        ****** unit scaling factor = 1 ******
######Streamout is finished!
```

7. Attach the snapshot of IR drop result. (**Make sure IR drop < 8%**) (5%)
   Open **ir_linear.gif** under the path …/PD_25C_avg_x/VDD and VSS. Take the snapshot.

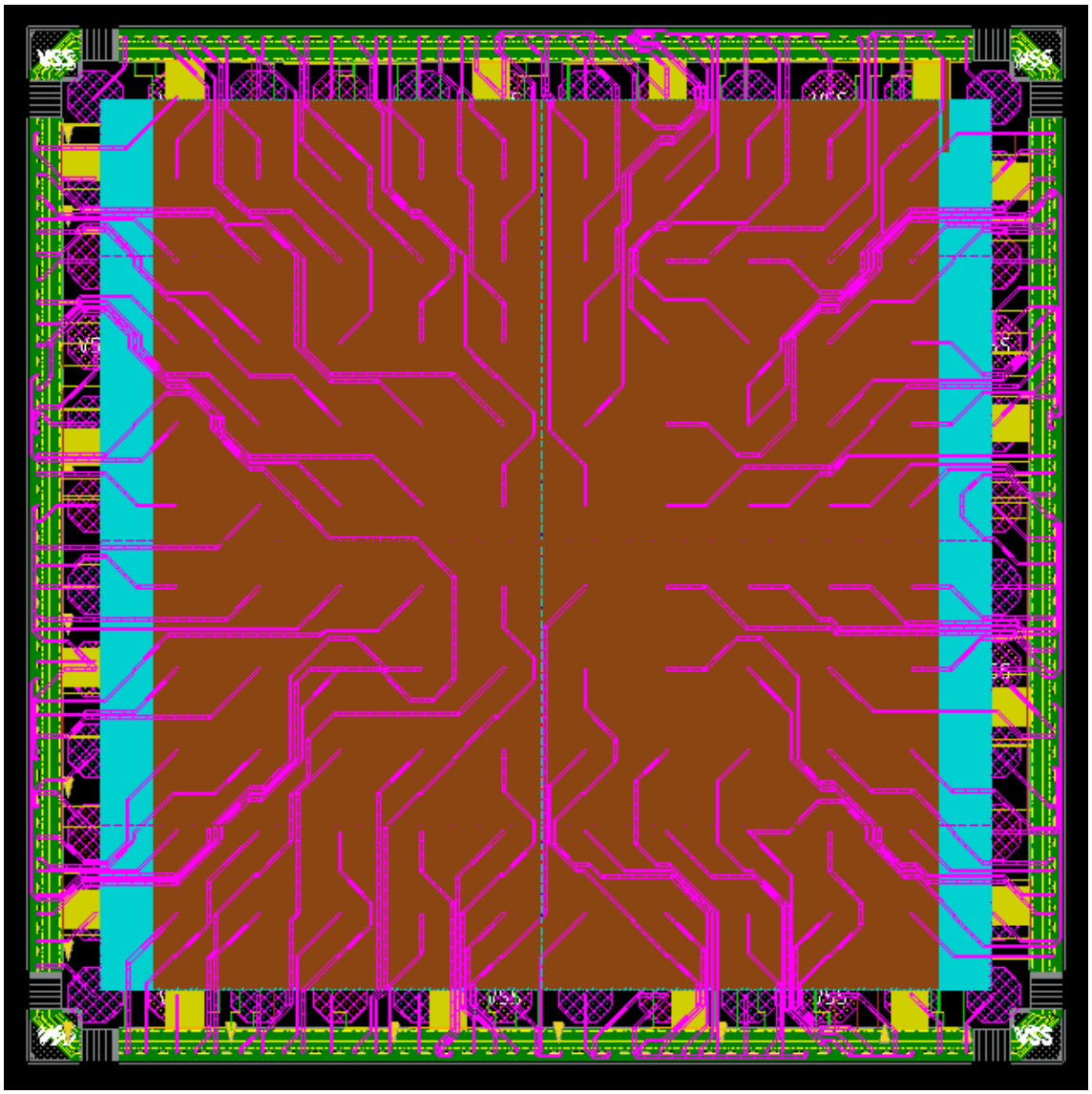




8. Attach the snapshot of the final area result. (5%)
   Through command：`analyzeFloorplan`
   (analyzeFloorplan destroys the design. Remember to save your design files first!!)

```
************************* Analyze Floorplan **************************
    Die Area(um^2)              : 2545090.57
    Core Area(um^2)             : 1475743.90
    Chip Density (Counting Std Cells and MACROs and IOs): 70.251%
    Core Density (Counting Std Cells and MACROs): 100.000%
    Average utilization         : 100.000%
    Number of instance(s)       : 512375
    Number of Macro(s)          : 0
    Number of IO Pin(s)         : 100
    Number of Power Domain(s) : 0
************************* Estimation Results *************************
**********************************************************************
```

9. Attach the snapshot of your final layout **after adding core filler**. (Remember to switch to **Physical view** and make Pin Shapes visible) (5%)



10. What is your strategy for bump assignment and for placing signal and power pins/pads? What is the reason behind it? (5%)

The bump assignment strategy was designed to balance signal integrity, power integrity, and routing efficiency. For power delivery, three IO power pads and four core power pads per side were distributed uniformly to provide stable power and reduce IR drop. Bumps were intentionally spread across the chip to avoid local current concentration and improve routability. In addition, power pads were placed on both sides of the clock signals to isolate them from other switching pads.